

BO1 :

Identify the few factors that truly influence the result, even when most available information is irrelevant.

DSO1:

Detect and prioritize the small subset of informative variables in datasets dominated by noise or irrelevant features.

BO2:

Understand and use the combined influence of several factors that only make sense when considered together.

DSO2:

Capture complex multivariate interactions that do not exhibit strong marginal effects but drive the underlying data structure.

BO3 :

Increase the reliability of decisions when dealing with situations that involve many pieces of information but very few examples to learn from.

DSO3:

Improve predictive performance in high-dimensional settings where the number of features greatly exceeds the number of observations ($p \gg n$).

Modèle	Variables impliquées	Paramètres
B01 + DSO1 Identifier les vraies variables utiles dans beaucoup de bruit	Variables fortes vs variables bruit	<i>Variable Importance (VI)</i> via ET <i>Variable Muting</i> (désactivation du bruit)
BO2 + DSO2 Capturer les interactions entre plusieurs variables	Variables interactives / combinées	<i>Linear Combination Split</i> <i>RL-based Variable Selection</i> (évalue gain futur)
BO3 + DSO3 Décider de façon fiable en high-dimension avec peu d'exemples	p très grand, n petit, vrai signal dans p_1 variables	<i>Ensemble RLT Convergence</i> dépendant de p_1 , <i>RL</i> pour robustesse

✓ DSO1 — Prioritizing Informative Variables

Category	Details
Models to Use	✓ RLT with variable muting enabled ✓ Optional checks: Random Forest or Lasso to confirm variable importance
Variables Involved	♦ Full feature set P at the start ♦ Muted set (Pd): irrelevant variables removed over rounds ♦ Protected set (P0): once selected, a variable is never removed ♦ Final strong variables = highest VI scores
Key Parameters	🔧 Muting rate: ~ 50% of remaining variables 🔧 Protected minimum (P0): 0–5 variables 🔧 Number of trees: ≈100 🔧 Minimum node size: 2 🔧 Embedded model: 100 trees, bootstrap ≈ 0.85

✓ DSO2 — Capturing Multivariate Interactions

Category	Details
Models to Use	✓ RLT with linear combination splits activated ✓ Reinforcement-based variable selection on ✓ Optional baselines: Gradient Boosting or BART
Variables Involved	◆ Top VI variables used to create interaction features ◆ Includes variables with no marginal effect, but strong joint effect ◆ Protected set P0 keeps key interaction variables ◆ Noise variables removed into Pd
Key Parameters	🔧 Max variables in linear combo (k): 2–5 🔧 VI threshold (α): ~0.25 🔧 Random split quantile (q): 0.1–0.5 🔧 Trees (M): ~100 🔧 Minimum node size (nmin): 2–5

✓ DSO3 — Improving Performance in $p \gg n$ Settings

Category	Details
Models to Use	✓ RLT with aggressive muting ✓ Optional comparison: ExtraTrees for speed
Variables Involved	◆ Full high-dimensional feature space P ◆ Small strong subset p_r detected early ◆ Muted set P_d grows quickly to remove noise ◆ Protected set P₀ stabilizes selection when n is small
Key Parameters	🔧 Muting rate: 70–80% of remaining variables 🔧 Protected minimum (P₀): usually not required 🔧 Minimum node size: 2

	🔧 Embedded model size: small/fast 🔧 Bootstrap ratio: ≈ 0.85 🔧 Trees: ≈ 100
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